

Algebraic Limits: Factoring and Rationalizing

Answer Key

AP Calculus AB/BC

Quick Answers

1. 6
2. -1
3. 12

4. $\frac{1}{4}$
5. $\frac{1}{6}$
6. $-\frac{1}{4}$

7. 9
8. —
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1. Direct substitution gives $\frac{0}{0}$. Factor the numerator as a difference of squares: $\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$ for $x \neq 3$. Substituting $x = 3$ gives $\boxed{6}$, which matches the table's approach from both sides.

2. Factor: $x^2 - 5x + 6 = (x - 2)(x - 3)$, so the quotient is $x - 3$ for $x \neq 2$. Substituting $x = 2$ gives $\boxed{-1}$.

3. Difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so the quotient is $x^2 + 2x + 4$. At $x = 2$ this is $4 + 4 + 4 = \boxed{12}$.

4. Multiply by the conjugate $\frac{\sqrt{x+4}+2}{\sqrt{x+4}-2}$. The numerator becomes $(x+4) - 4 = x$, giving $\frac{x}{x(\sqrt{x+4}-2)} = \frac{1}{\sqrt{x+4}-2}$. At $x = 0$ this is $\frac{1}{2-2} = \boxed{1/4}$.

5. Multiply by $\frac{\sqrt{x}+3}{\sqrt{x}+3}$. The numerator becomes $x - 9$, so the quotient is $\frac{1}{\sqrt{x}+3}$. At $x = 9$ this is $\frac{1}{3+3} = \boxed{1/6}$.

6. Multiply by $\frac{2 + \sqrt{x+3}}{2 + \sqrt{x+3}}$. The numerator becomes $4 - (x+3) = 1 - x = -(x-1)$, so the quotient is $\frac{-1}{2 + \sqrt{x+3}}$. At $x = 1$ this is $\frac{-1}{2+2} = \boxed{-1/4}$. The negative sign comes from the radical being subtracted, not from an algebra slip.

7. For $x \neq 7$, $\frac{x^2 - 5x - 14}{x - 7} = \frac{(x - 7)(x + 2)}{x - 7} = x + 2$, so $\lim_{x \rightarrow 7} f(x) = 9$. Continuity at $x = 7$ requires $f(7)$ to equal that limit, so $c = \boxed{9}$.

8. Manual review — expected reasoning. A limit describes the behaviour of f as x approaches 3, and the definition of the limit never evaluates f at 3 itself. For every $x \neq 3$ the two expressions agree, so they have the same limit. At $x = 3$ they differ: $x + 3$ is defined and equals 6, while $\frac{x^2 - 9}{x - 3}$ is undefined — a removable discontinuity. Accept any answer making both points: agreement away from the point, and the limit not depending on the value at the point. Do not accept “the 3s cancel.”